

Chemistry Quiz Prep: Bonding & Ions

Topics: Metallic Bonds and Alloys · Ionic Bonding · Polyatomic Ions · Monoatomic Ions **Source:** *Ch9 — Atomic Foundations of Matter* (sections 9.4.2, 9.5, 9.6, Table 9.1) and *Ch5 — Exploring Mixtures and their Separation* (alloys). One part is supplemented from standard curriculum knowledge — flagged clearly where it appears, since it wasn't found in either chapter.

How to use this: This is a fast, quiz-ready review, not a full lesson — read it once slowly, then skim it again right before the test. Flashcards for quick self-testing are linked separately.

1. Metallic Bonds and Alloys

Metallic bonding (not in your two chapters — standard chemistry model, included so this topic is covered)

Metal atoms in a solid release their loosely-held outer (valence) electrons into a shared “sea” that moves freely around the whole structure. What’s left behind is a lattice of positively charged metal ions sitting inside this sea of mobile (“delocalized”) electrons.

Metallic bond = the electrostatic attraction between the positive metal ions and the surrounding sea of delocalized electrons.

This one model explains all the typical metal properties:

Property	Why (from the electron-sea model)
Good conductor of electricity & heat	Free electrons carry charge/energy easily
Malleable (hammers into sheets)	Layers of ions slide past each other; electron sea just flows around them, bond doesn't break
Ductile (draws into wires)	Same reason as malleability
Lustrous (shiny)	Free electrons reflect light
Generally high melting/boiling point	Strong attraction between ions and electron sea

Alloys (this part is in your Ch5 textbook)

- Metals do **not** dissolve into each other at room temperature.
- Melt two or more metals at high temperature → they mix into a solution → cool it → it solidifies into a new material that looks like a single metal.
- Definition:** an alloy is a **homogeneous mixture** of two or more metals, or a metal and a non-metal.
- Alloys **cannot** be separated by physical methods.
- Why make alloys?** To get materials that are stronger, more rigid, or more corrosion-resistant than the pure metal.
- An alloy is a **mixture, not a compound** — no new chemical bond forms between the different metal atoms; it's physical mixing at the atomic scale.

Memorize these three (straight from the textbook):

Alloy	Composition
Brass	~80% copper + ~20% zinc

Bronze	~80% copper + ~20% tin
Stainless steel	Iron + carbon (0.03-0.8%) + chromium (16-18%) + nickel (10-14%) + molybdenum (2-3%)

2. Ionic Bonding

Rule of thumb: fewer than 4 valence electrons → atom **donates/loses** electrons → forms a **positive** ion. More than 4 valence electrons → atom usually **gains or shares** electrons to complete its octet.

Ionic bond = the electrostatic force of attraction between oppositely charged ions that holds them together.

Formed by **transfer of electrons**, typically from a metal atom to a non-metal atom.

Classic example: sodium chloride (NaCl)

- Sodium (Na): 1 valence electron → **loses it** → becomes **Na⁺** (sodium cation) — 11 protons, 10 electrons.
- Chlorine (Cl): 7 valence electrons → **gains 1 electron** → becomes **Cl⁻** (chloride anion).
- Na⁺ and Cl⁻ attract each other electrostatically → ionic bond → NaCl.

Vocabulary:

- Cation** = positively charged ion (atom **lost** electrons).
- Anion** = negatively charged ion (atom **gained** electrons).
- Cations + anions, together = “ions.”

Ionic compounds form 3-D crystal lattices, not molecules. In NaCl, each Na⁺ is surrounded by 6 Cl⁻, and each Cl⁻ is surrounded by 6 Na⁺, in a repeating pattern (a crystal lattice).

Naming ionic compounds

- Cation name first, then anion name.
- Simple anion names end in “**-ide**” (chloride, oxide, sulfide...).
- Metals form cations; non-metals form anions.
- Polyatomic ion names generally do **not** end in “-ide” (see Section 3).

Writing formulas — the criss-cross method

- Write the cation symbol first, then the anion symbol.
- Write the charges under the symbols (not as superscripts).
- Criss-cross the numbers (only) to get the subscripts.
- Simplify the subscripts by their common factor, if any.

Worked examples:

Compound	Cation	Anion	Formula
Calcium chloride	Ca ²⁺	Cl ⁻	CaCl ₂
Aluminium oxide	Al ³⁺	O ²⁻	Al ₂ O ₃
Magnesium oxide	Mg ²⁺	O ²⁻	MgO (2:2 simplifies to 1:1)
Calcium carbonate	Ca ²⁺	CO ₃ ²⁻	CaCO ₃ (2:2 simplifies to 1:1)

Magnesium hydroxide	Mg ²⁺	OH ⁻	Mg(OH) ₂
Aluminium hydroxide	Al ³⁺	OH ⁻	Al(OH) ₃
Aluminium sulfate	Al ³⁺	SO ₄ ²⁻	Al ₂ (SO ₄) ₃

Brackets () are only needed when **2 or more of the same polyatomic ion** appear in the formula.

Properties of ionic compounds

Property	Ionic compounds
Solubility	Soluble in water; insoluble in kerosene/petrol
Conducts electricity as a solid?	No — ions are locked in the lattice, can't move
Conducts electricity molten or dissolved?	Yes — ions become free to move
Melting / boiling point	High — strong inter-ionic attraction

3. Polyatomic Ions

Definition: ions made of **two or more atoms bonded together**, carrying one overall charge. Names generally do **not** end in “-ide.”

Name of ion	Formula	Valency
Hydroxide	OH ⁻	1
Nitrate	NO ₃ ⁻	1
Hydrogencarbonate (bicarbonate)	HCO ₃ ⁻	1
Carbonate	CO ₃ ²⁻	2
Sulfate	SO ₄ ²⁻	2
Ammonium	NH ₄ ⁺	1

Watch out: Ammonium (NH₄⁺) is the only **cation** in this table — every other polyatomic ion here is an anion.

4. Monoatomic Ions

Definition: an ion made from a **single atom** that has gained or lost electrons (contrast with polyatomic ions, made of multiple bonded atoms).

Cations (metals — lose electrons):

Name of ion	Formula	Valency
Sodium	Na ⁺	1
Lithium	Li ⁺	1
Potassium	K ⁺	1
Silver	Ag ⁺	1
Calcium	Ca ²⁺	2
Barium	Ba ²⁺	2

Iron (Ferrous)	Fe^{2+}	2
Iron (Ferric)	Fe^{3+}	3
Copper (Cuprous)	Cu^{+}	1
Copper (Cupric)	Cu^{2+}	2
Magnesium	Mg^{2+}	2
Zinc	Zn^{2+}	2
Aluminium	Al^{3+}	3

Anions (non-metals — gain electrons):

Name of ion	Formula	Valency
Fluoride	F^{-}	1
Chloride	Cl^{-}	1
Bromide	Br^{-}	1
Iodide	I^{-}	1
Oxide	O^{2-}	2
Sulfide	S^{2-}	2

Trickiest pair to double-check: iron and copper each have **two** possible ions — know which name goes with which charge:

- Ferrous = Fe^{2+} · Ferric = Fe^{3+}
- Cuprous = Cu^{+} · Cupric = Cu^{2+}

60-Second Self-Check

Cover the right-hand column and test yourself.

Question	Answer
What holds a metal lattice together?	Electrostatic attraction between metal cations and a “sea” of delocalized electrons
Why are metals malleable?	Ion layers slide past each other while the electron sea keeps the bond intact
What is an alloy?	A homogeneous mixture of two or more metals, or a metal and a non-metal
Composition of brass?	~80% copper, ~20% zinc
What is an ionic bond?	Electrostatic attraction between oppositely charged ions
How does Na become Na^{+} ?	It loses its one valence electron
Do ionic solids conduct electricity?	No — only when molten or dissolved in water
What’s a polyatomic ion?	An ion made of 2+ atoms bonded together, carrying one overall charge
Name the only common polyatomic cation	Ammonium, NH_4^{+}

What's a monoatomic ion?	An ion formed from a single atom
Ferric vs. Ferrous?	Ferric = Fe^{3+} , Ferrous = Fe^{2+}

Notes prepared from Ch9 "Atomic Foundations of Matter" and Ch5 "Exploring Mixtures and their Separation." The metallic-bonding electron-sea model was not found in either chapter and is included from standard chemistry curriculum content, since "metallic bonds" was one of the requested topics. If your teacher covered metallic bonding differently in class, check your class notes against Section 1 before the test.